The Extraction Skill: Winning Contact Is a Persistent, Individual, and Unrewarded Talent Across Six Leagues

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Introduction

Every drawn penalty hands over a power play, yet hockey records the event only against the offender — the player who *caused* it receives no statistic at all. We ask whether drawing penalties is a persistent individual skill, what it is worth in wins, and whether anyone pays for it. The answer matters directly to roster construction: we find a repeatable talent worth roughly **a third of a win per season** that is uncorrelated with scoring and invisible to conventional valuation.

Methods

Using NHL play-by-play (2023-24 to 2025-26; 3,936 games; 141,711 player-games), we model even-strength minors drawn as a weighted Poisson process with contextual covariates (position, venue, leave-one-out opponent discipline, score state, team environment) and a Gamma-Poisson empirical-Bayes skill multiplier. We validate strictly out of sample (fit on two seasons, predict a third), replicate across fifteen seasons, and run three isolation tests: an event-level ridge design over 20,369 penalties separating individual extraction from on-ice penalty tilt; team-environment controls; and a disjoint-referee-crew reliability test built from a full officials census. Value is estimated empirically against same-state (score, time, venue) win-rate baselines and triangulated through goal conversion. The identical construction is then applied to the NBA, WNBA, both NCAA basketball leagues (sportsdataverse), and international soccer (StatsBomb).

Results

Out of sample, adding the skill layer raises predictive correlation for player draw totals from **0.25** (exposure) and **0.53** (context) to **0.72**; held-out draw rates are monotone in prior skill quartiles (0.41→0.81 per hour). Persistence is positive in **14 of 14** adjacent season pairs since 2010 (mean r=0.59; career reliability 0.89). Extraction is distinct from possession-driven penalty tilt (r=0.087) and survives disjoint referee crews (r=0.51 vs. 0.49 random-split calibration). A drawn minor is worth **+0.017 win probability** (goals path: +0.118 goals); elite net drawers add ~0.3 wins/season, and roster-level spread reaches ~2.7 wins.

The skill decomposes into two stable stylistic routes — provocation vs. puck-carrier extraction — with a speed mechanism confirmed two independent ways: differential within-player aging (-2.6% vs. -4.0%/yr), and NHL Edge tracking data, where measured skating speed predicts carrier-route draws (partial r=0.17–0.23, position- and possession-controlled) but not provocation draws (r=0.04). It replicates in all five additional leagues (reliability 0.71–0.94). It correlates with points/60 at **0.067**, and deployment runs mildly against it (-0.28 min/game per SD conditional on scoring; -0.18 and still significant after NHL Edge zone-start deployment controls).

Conclusion

Winning contact is a measurable, career-stable, win-valued talent that hockey's accounting renders structurally invisible: unlike basketball, where drawn fouls surface as free-throw points, a drawn penalty never touches the drawer's ledger. Teams can acquire penalty differential — prior-season roster skill predicts it at r=0.45 — at a price of approximately zero.

> Code, derived data, and full methodology: **github.com/Murray22/extraction-skill** · Platform: gamevibeanalytics.com

Figure 1 · One construction, six leagues

LEAGUE	CORPUS	RELIABILITY (SB)	YEAR-OVER-YEAR	TOP OF BOARD
NHL (core)	3 seasons · 3,936 games	0.89 (career)	0.59 mean × 14 pairs	Hathaway 2.45×, M. Tkachuk, Kadri
NBA	5 seasons · 5,830 games	0.94	0.86–0.88 × 4	Antetokounmpo 3.90×, Williamson, Dončić
WNBA	6 seasons · 1,306 games	0.86	0.72–0.82 × 5	Wilson 2.34×, Stewart, Reese
NCAA M	10 seasons · ~1.2M trips	—	0.54–0.66 × 9	Dybantsa 2.16×
NCAA W	11 seasons	—	0.59–0.67 × 8	Nichols 2.63×
Soccer	8 tournaments · 1,179 matches	0.71	—	Hazard in pooled top 10

Figure 1. The identical three-step construction (contextual expectation → shrunken skill ratio → persistence test) applied to raw event data in six leagues. Basketball drawn-by attribution is structurally inferred via foul→free-throw linkage.

Figure 2 · Out-of-sample validation

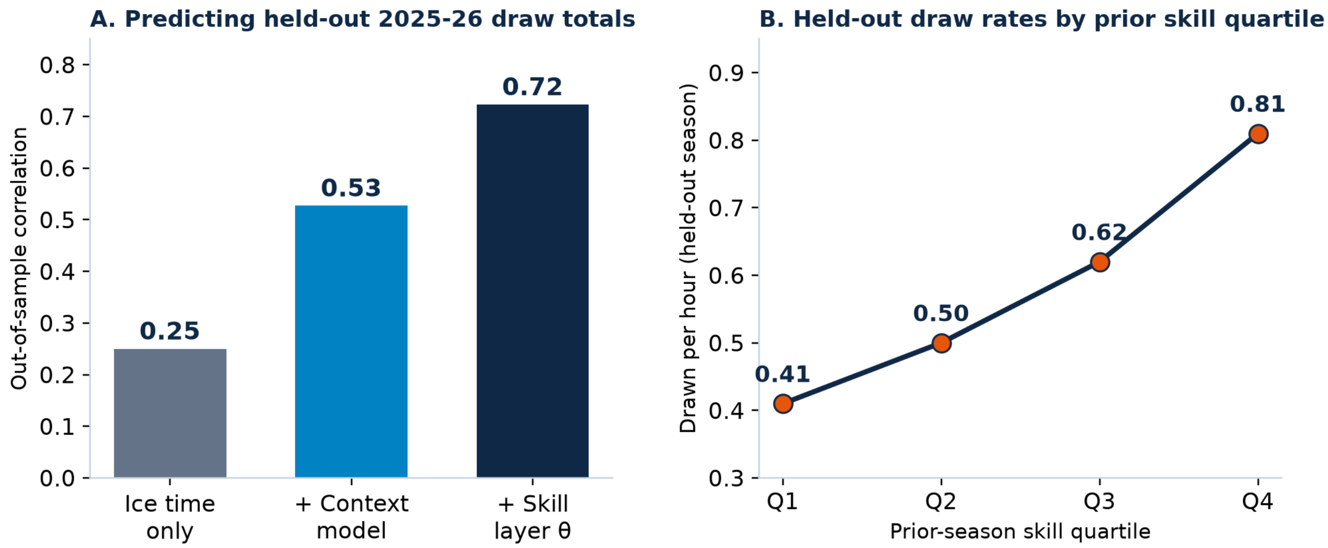

Figure 2. (A) Predictive correlation for held-out 2025-26 player draw totals as model layers are added. (B) Raw held-out draw rates by skill quartile fit on prior seasons only — the top quartile draws at twice the rate of the bottom in a season the model never saw.